BELLWETHER-2 — 3-Qubit GHZ Mermin Multipartite Witness Report

Experiment Design

3-qubit GHZ Mermin-3 inequality test bound into an authorization ProofRecord as a multipartite spatial nonclassicality witness. Executed on IBM Quantum backend ibm_fez, job d9dno91htsac739da4d0 (Remnant Fieldworks Inc.).

Measured Results

- M = -3.423
- E_XYY = -0.888, E_YXY = -0.824, E_YYX = -0.831, E_XXX = 0.88
- sigma = 0.023103
- shots_per_setting = 2000

Verdict

Significant multipartite-nonclassicality violation UNDER STATED TRUST ASSUMPTIONS

Honesty Boundary

This is a device-dependent multipartite witness on co-located superconducting qubits. Locality loophole OPEN; detection loophole OPEN. This result is NOT loophole-free and NOT device-independent certified randomness. Classical bound: $|M| \leq 2$. Quantum (GHZ) maximum: $|M| = 4$.

Provenance

- mermin_nonce: 20f8a53a7dac75875ece12131a2dbc7c431ed5a1f48cc1e01d34b996c10832ac
- record_hash: b69903f1e48342fb68ff2f832c1be53e3e1d4a2a8859aa1c4db66e025fa0fbcd

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Zenodo published record: https://zenodo.org/records/21430446
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To God be the glory. Proof Before Power. Verification Before Execution.